AIAA 3201 · PROJECT 4 · PART 1

# Initialization, not architecture

COLMAP vs. VGGT-1B vs. VGGT-Ω as 3D Gaussian Splatting initialization under a controlled dense view budget

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# What we compared, and how

FOUR AXES, AS SPECIFIED IN THE BRIEF

**Initialization runtime**
seconds to a training-ready scene

**Pose accuracy**
ATE after Sim(3) alignment; relative rotation

**3DGS convergence speed**
test PSNR every 5 k iterations to 30 k

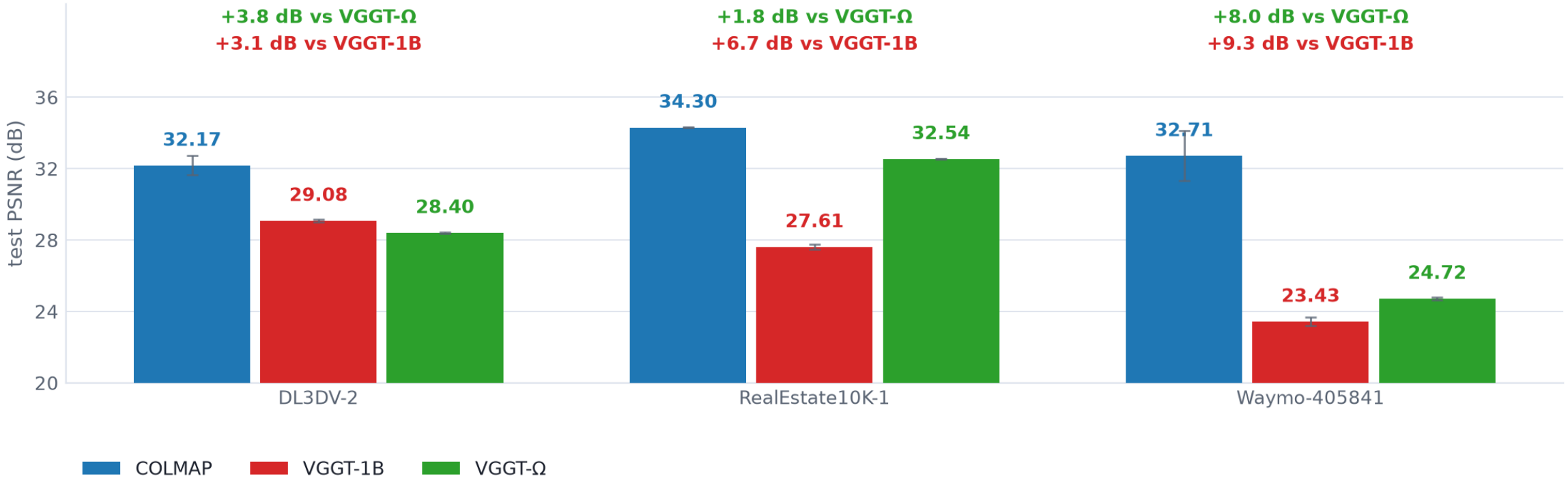
**Final rendering quality**
PSNR / SSIM / LPIPS on frozen test views

THREE MANDATORY SCENES

| Scene | Type | Images | Test |
|---|---|---|---|
| **DL3DV-2** | outdoor | 306 | 39 |
| **RealEstate10K-1** | indoor | 279 | 35 |
| **Waymo-405841** | driving | 199 | 25 |

- N = 160 training views — the dense regime; 8–17× denser than Part 2's sparse settings
- Splits frozen, llffhold = 8 — all three backends see the same images
- 30 000 iterations, densify to 15 000, reference hyper-parameters
- 2–3 seeds per configuration, with pairwise t-tests

# COLMAP produces the best renders on all three scenes




*y-axis truncated at 20 dB · error bars = run-to-run std (2–3 seeds)*

**Largest gap on Waymo-405841 (+9.3 dB).** SSIM and LPIPS rank the backends identically, so the ordering is not an artefact of one metric.

# The ordering is visible without metrics




*Column headers give the scene-level mean test PSNR (mean over all test views, 2–3 seeds where available); corner labels give that single view's PSNR.*
*Note: the official VGGT exporter caps the point cloud at 100 000 points chosen by a random unseeded subsample (vggt/utils/helper.py::randomly_limit_trues), so VGGT-1B figures carry an export-level spread far larger than its training-seed spread — see the report's limitations.*

## WHAT TO LOOK AT

**VGGT-1B, 21.4 dB**

loses the fine structure of the red foliage and the paving joints.

**VGGT-Ω, 28.3 dB**

keeps the shape but drifts in colour — visibly greyer and colder.

**COLMAP, 32.7 dB**

closest to ground truth on all three rows.

## ONE THING TO BE CAREFUL ABOUT

These renders use a different VGGT-1B export than the 29.1 dB in the table. The exporter subsamples the cloud with an unseeded **np.random.choice**, so that column moves by several dB between exports of the same model.

# The mechanism is not pose accuracy

POINT CLOUD, DL3DV-2 (model_analyzer)

| Model | Points | Mean track length |
|---|---|---|
| **COLMAP** | 3 778 | **8.19** |
| **VGGT-1B** | 100 000 | 1.000 |
| **VGGT-Ω** | 150 000 | 1.000 |

TRACK LENGTH = 1.000 MEANS EVERY 3D POINT IS SEEN IN EXACTLY ONE IMAGE

— Both exporters emit per-frame depth unprojections, not multi-view triangulations.

— 3DGS therefore starts from mutually unconstrained samples and must reconcile their disagreements at render time.

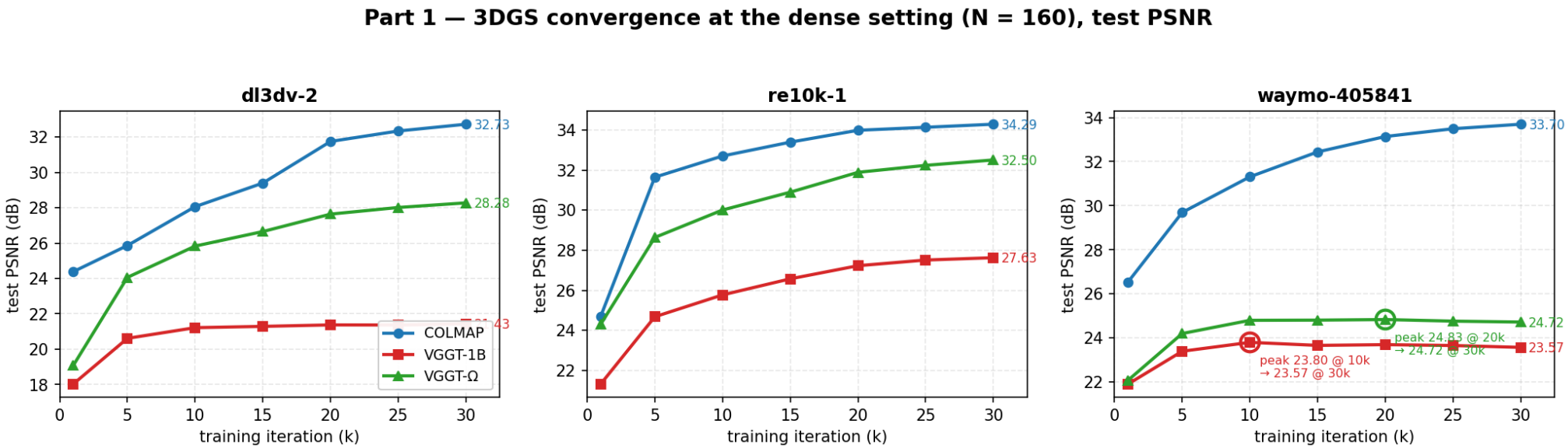
— The reported 0.000 px reprojection error is vacuous: a point seen once has no cross-view residual.

> **The contradiction, at 50 views (Re10k-1):**
> VGGT-Ω has **5× COLMAP's pose accuracy** (ATE 0.220 % vs 1.065 %) and still renders **1.35 dB worse**.

SO WHAT SHOULD YOU CONCLUDE?

- Pose accuracy alone cannot explain rendering quality.
- The feed-forward models are not worse reconstructors — they deliver competitive poses and dense but unconstrained geometry.
- Follow-up we did not run: keep VGGT's poses, re-triangulate points from multi-view correspondences.

# Convergence: 30 k is the right budget at 160 views



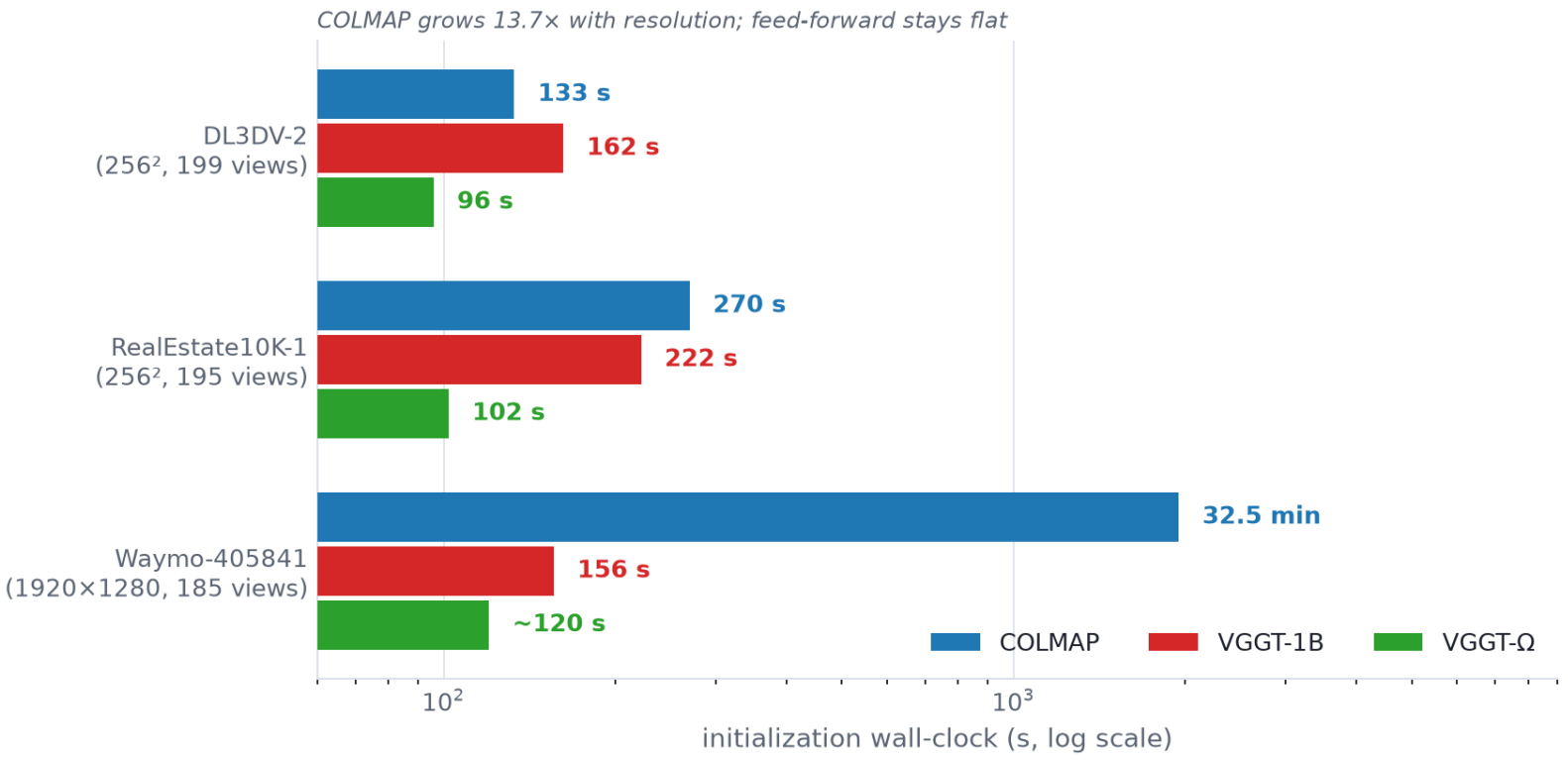

*Hollow circles mark configurations whose peak occurs before 30 k and then decays; the two numbers give peak and final test PSNR.*

**7 of 9 configurations** are still improving at 30 k — and all three COLMAP curves are.

**Waymo is the exception:** both feed-forward models peak early (10 k, 20 k) and then decay; COLMAP does not.

**The "peak at 1 k" from the N=50 run** does not occur at 160 — it was under-training, not over-fitting.

# Speed and run-to-run stability




READING THE CHART

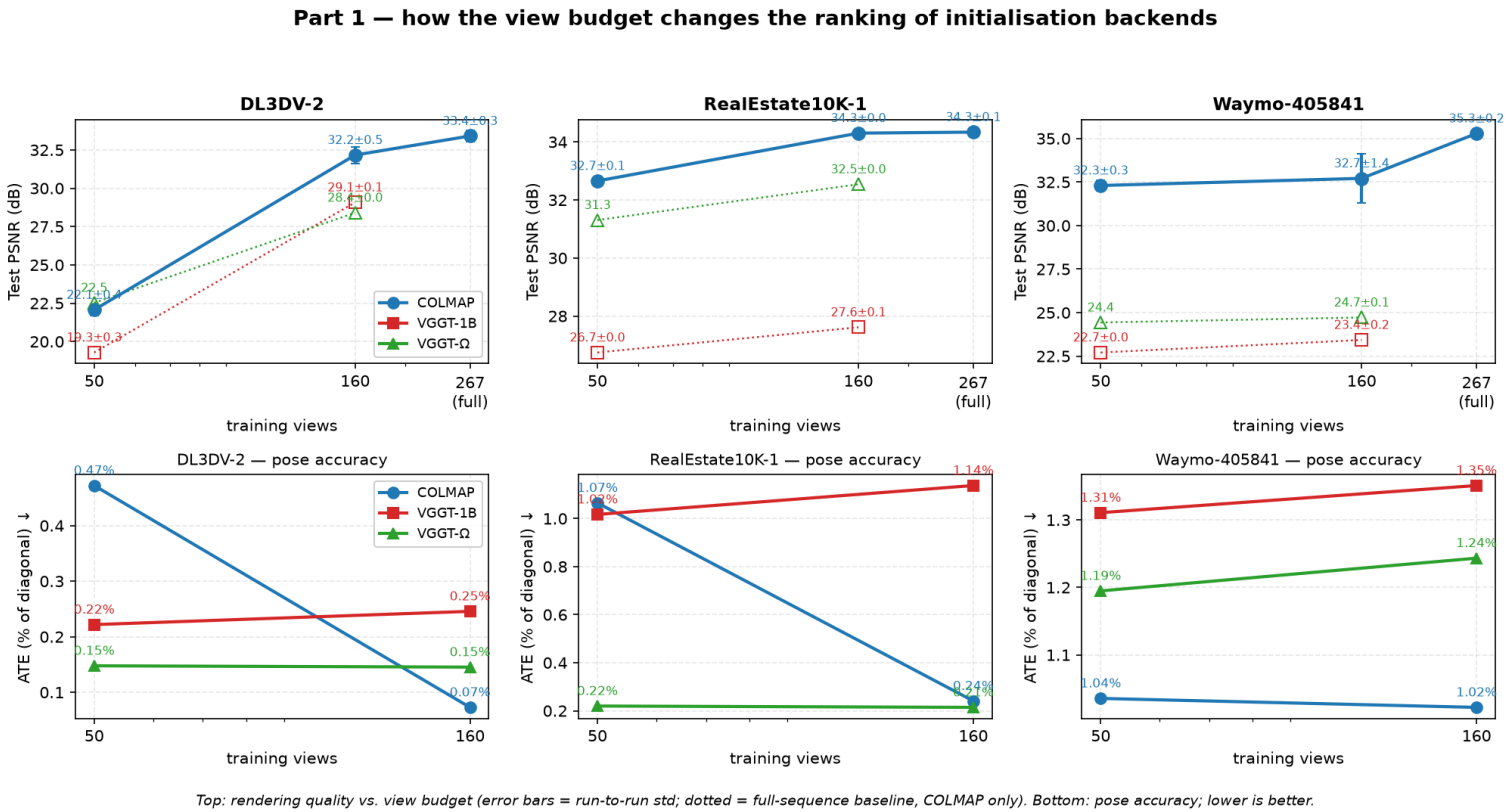
— Feed-forward: 70–220 s regardless of input resolution (every input is normalized to a fixed 512² window).

— COLMAP: 133 s → 1 950 s as resolution grows from 256² to 1920×1280.

AND IT REPRODUCES

- **8 of 9 configurations** reproduced to within 0.12 dB on a second GPU, rebuilt from scratch.
- **The one outlier:** VGGT-1B's exporter subsamples the cloud with an unseeded np.random.choice.

# The gap widens as observations accumulate




*Top: rendering quality vs. view budget (error bars = run-to-run std; dotted = full-sequence baseline, COLMAP only). Bottom: pose accuracy; lower is better.*

**COLMAP's pose error collapses 6.5× (DL3DV-2) and 4.4× (Re10k-1) from 50 → 160 views; feed-forward error does not move.** VGGT-Ω gains the least (+5.85 dB vs COLMAP's +10.03 on DL3DV-2), which is what moves it from first place at 50 views to last at 160.

# Bonus: it also holds on Mip-NeRF 360

THE 3DGS PAPER'S OWN BENCHMARK — OPTIONAL IN THE BRIEF

| Scene | Images | COLMAP | VGGT-1B | Gap |
|---|---|---|---|---|
| **counter** | 240 | **29.47** | 23.66 | +5.81 |
| **bonsai** | 292 | **31.27** | 17.03 | **+14.24** |
| **flowers** | 173 | **21.49** | 15.51 | +5.99 |

**COLMAP is re-run by us on the same frames**, not the bundled all-frame model — otherwise the comparison would not be like-for-like.

WHAT IT SHOWS

**+5.8 to +14.2 dB.** The COLMAP advantage is the same order as on the three mandatory scenes, on a benchmark we did not tune on.

**bonsai is the strongest form of the scaling asymmetry.** Going from 50 to 160 views, COLMAP improves (28.33 → 31.27 dB) while VGGT-1B gets **worse** (19.05 → 17.03 dB).

**Caveats.** Six configurations, n = 1, no significance test. VGGT-Ω was not run here. kitchen is absent from the weights mirror we used, so flowers was substituted.

**3 of 3 scenes, same direction, no reversal —** the optional-deliverable check passes, and on a benchmark we never tuned on.

# Seven silent failure modes

**Every number above required eliminating these first.** None produced an error message. Two individually exceeded the effects we report as findings.

| # | Failure | Silent symptom | Impact |
|---|---|---|---|
| 1 | Diverged COLMAP camera | camera centre at $7.2\times10^{9}$; scene box $7.2\times10^{9}$ | **+13.0 dB** |
| 2 | Double-normalized focal in the Ω export | exported fx = 0.4 instead of 204 | **+13.4 dB** |
| 3 | Mixed alignment procedures | rotation error inflated 20× on a near-planar path | would mis-rank |
| 4 | Wrong Waymo GT convention | rotation error 23.5° instead of 1.38° | poses look broken |
| 5 | Dataset poses hide SfM cost | initialization time = 0 | speed axis impossible |
| 6 | Missing sparse/0 directory | "Could not recognize scene type!" | pipeline won't start |
| 7 | LPIPS weights fetched at eval time | 528 MB at 150 KB/s | ~1 h of stalls |

**Failures 1 and 2 are the same species:** scale errors that leave the optimizer running happily, detectable only by checking an intermediate quantity against a known reference. Watching the loss curve would not have caught either.

# Takeaways

> **COLMAP-initialized 3DGS renders best on all three scenes, and the margin widens as observations accumulate.**

- **Pose accuracy is not the explanation.** COLMAP's error falls 4–6× from 50 → 160 views while feed-forward error stays flat — and at 50 views COLMAP has the worst poses yet the best renders.
- **VGGT-Ω is the best feed-forward option.** By far the most reproducible (std 0.03–0.07 dB), and it beats COLMAP at 50 views on DL3DV-2 — but it gains least from more views and falls to last place.
- **The decisive variable is point-cloud topology.** Track length 8.19 for triangulation vs 1.000 for per-frame depth unprojection.
- **What data-driven scaling buys:** a 70–100 s, resolution-insensitive, highly reproducible initializer — at the cost of 3–9 dB of fidelity.

**Limitations:** VGGT-1B's export is stochastic (±several dB); Mip-NeRF 360 is single-seed; no hybrid VGGT-poses + re-triangulated-points run; COLMAP timings are CPU-bound.